

Dong Su Kim

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WORK EXPERIENCE

Conboy Lab

Machine Learning Engineer Intern

October 2025

- Designed a **scalable distributed data pipeline** for multi-terabyte datasets, improving preprocessing speed **30%** through parallelization, optimized I/O, and batching.
- Developed end-to-end ML workflows including feature extraction, model training, evaluation, and automated retraining hooks to maintain **98% uptime** for inference pipelines.
- Productionized ML pipelines using **Docker, CI/CD, and experiment versioning**, reducing configuration failures by **~25%** across cloud and HPC environments.
- Implemented **model-serving infrastructure** with FastAPI and AWS (S3, EC2) to support async inference and multi-stage preprocessing.

PROJECTS

Sequence Modeling & ML Inference Pipeline

End-to-end system for protein sequence inference.

- Designed and deployed a **Dockerized ML** inference service integrating BERT/ESM2 embeddings with async **FastAPI** endpoints and cloud-backed storage.
- Built a **microservice pipeline** with preprocessing, embedding generation, model inference, and results aggregation supporting over **~50,000** sequences per run.
- Integrated **Postgres + S3** for structured metadata and bulk sequence storage, reduced data access latency by **35%**.
- Improved predictive performance **15%** through custom data augmentation and systematic hyperparameter tuning.

Ambra

Agentic EMS Ambient Capture

- Built an end-to-end EMS **agentic AI pipeline** that converts raw ambulance audio into fully structured claims exporting to **7+ industry standard** documentation formats (NEMSIS v3.5, X12 837P, FHIR R4, coverage reports).
- Engineered **ASR + diarization + clinical NLP** stack extracting **50+ structured EMS fields** (vitals, meds, procedures, patient data) with schema-validated outputs.
- Designed and implemented a **modular agent orchestration workflow** with session state management, reducing end-to-end processing time by **>40%** through **parallel stage execution** and reusable agent components.
- Developed a production-grade **CLI + test suite with 80+ automated tests**, ensuring reproducible builds and **100% format compliance** across all export targets.

Hunger

Contextually Scalable Restaurant Rating Tracker - Beli Clone

- Built a full-stack restaurant discovery app indexing **1,000+ restaurants** with geospatial filtering and **<150ms** query latency via optimized geocoding and caching.
- Designed an ML-guided recommendation engine reducing **500+ candidates to top 1-3 matches** using a questionnaire, heuristics, and XGBoost ranking across **20+ features**.
- Developed a synthetic persona generator producing **10,000+ user profiles** with weighted cuisine distributions to enable offline evaluation of recommendation quality.
- Engineered **15+ modular async FastAPI endpoints** with reusable components and client-side validation, documented in a detailed design spec for maintainability.

EDUCATION

UC Berkeley

Computer Science, Bioengineering

June 2024

Berkeley, CA

SKILLS

Programming: Python, SQL, Java, C, JavaScript/TypeScript/React, CSS, html

ML & DL: PyTorch, TensorFlow, Keras, Hugging Face, RAG, MCP

ML Engineering: Model deployment, experiment tracking, distributed data processing, evaluation pipelines, GPU optimization

Data & Infra: Postgres, FastAPI, Docker, AWS (EC2, S3), CI/CD, Linux, HPC

Valid EAD via EB-4 Visa - No Sponsorship Required